

Examination Multivariate Statistical Methods

Linköpings Universitet, IDA, Statistik

Course code and name:	732A83 Statistical Methods
Date:	2025/12/11, 8–12
Examinator:	Krzysztof Bartoszek
Allowed aids:	Pocket calculator Table with common formulae and moment generating functions (distributed with the exam) Table of integrals (distributed with the exam) Table with distributions from Appendix in the course book (distributed with the exam) One double sided A4 page with own hand written notes
Grades:	A= $[18 - \infty)$ points B= $[16 - 18)$ points C= $[14 - 16)$ points D= $[12 - 14)$ points E= $[10 - 12)$ points F= $[0 - 10)$ points
Instructions:	Write clear and concise answers to the questions. Reporting on the exam a probability outside the $[0, 1]$ interval or a negative variance without a comment that a mistake must have been made somewhere will result in a NEGATIVE point for each such instance.

Problem 1 (3p)

A certain DNA sequencer reads a DNA sequence without any error on any position with probability 0.1. Three separate DNA sequences are read independently by three separate (of the above type) DNA sequencers.

1. (1p) What is the probability that all three sequences are read correctly?
2. (1p) What is the probability that at least one sequence is read incorrectly?
3. (1p) What is the probability that at exactly one sequence is read incorrectly?

Problem 2 (5p)

Another DNA sequencer reads each position correctly with probability 0.9. To minimize the errors each DNA sequence is read three times and then the majority rule is used, i.e., if a given nucleotide (a letter from the set $\{A, C, G, T\}$) is present at the position in two or three of the reads, then it is chosen. Otherwise an N (unknown) is reported. A three nucleotide sequence is provided to the sequencer. For example if the true sequence is AGT and the three reads return $\{AGT, AGC, ACA\}$ then it will be reported as AGN .

1. (1p) What is the probability that a given position is identified correctly?
2. (1p) What is the probability that a given position is identified incorrectly?
3. (1p) What is the probability that a given position is impossible to identify?
4. (1p) What is the probability that the three nucleotide sequence is identified correctly?
5. (1p) What is the probability that the three nucleotide sequence is identified incorrectly (i.e., there is a wrongly identified position somewhere in the returned sequence)?

Problem 3 (3p)

The time to failure in years of a printer drum is a continuous random variable with density function

$$f(x) = \begin{cases} C(x^{-3} + x^{-7}) & x \geq 2, \\ 0 & x < 2. \end{cases}$$

1. (1p) What is the earliest possible time of failure?
2. (1p) Calculate the constant C and find the cumulative distribution function of the lifetime.
3. (1p) Find the expectation and variance of the time to failure.

Problem 4 (5p)

Let $X_1, \dots, X_n$ be independent and identically distributed from the Rayleigh distribution with unknown parameter θ :

$$f_{\theta}(x) = \begin{cases} \theta x e^{-\theta x^2/2}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

1. (3p) Find the maximum likelihood estimator of θ .
2. (1p) Calculate the maximum likelihood estimate of θ for the sample 1, 2, 2, 3.
3. (1p) It is known that the expectation of a Rayleigh distribution equals $\sqrt{\pi/(2\theta)}$. Calculate the method of moment estimator of θ for the above sample. If needed you may use that $\pi \approx 3.14$.

Problem 5 (4p)

The number of buses arriving at a bus stop every minute follows a Poisson distribution with mean λ . While waiting for your bus you decide to use Bayesian statistics to make inference on λ , with the prior being the prior $\lambda \sim \Gamma(a, b)$.

1. (2p) Find the posterior distribution of λ , and an expression for the Bayesian estimator, $\hat{\lambda}$.
2. (1p) Suppose that during your waiting time of 2 minutes and you counted a total of 10 buses. What would be the estimate for the average number of buses per minute, if the $\Gamma(2, 2)$ distribution is used for the prior?
3. (1p) When you wait for a bus you usually wait for a short amount of time, e.g., 5–20 minutes. Would the Poisson model be sensible if you stood at the bus stop and counted buses for a long time, e.g., a full day? Motivate your answer.